Outline of the topic and research project

The geometry of subvarieties on abelian varieties

The central question of this doctoral project is how much the tangent spaces of a subvariety determine its geometry inside a complex abelian variety. Translation identifies all ambient tangent spaces with a fixed vector space, giving the tangent Gauss map. For a curve, this map assigns its tangent direction to each smooth point. Understanding its degree and ramification is closely related to understanding the monodromy of the conormal Gauss map and, through characteristic cycles, the representation theory associated with the curve.

The initial motivation was the expectation that a failure of generic injectivity should be explained by ambient symmetries, such as translations or reflections. Preliminary work investigated this expectation for curves in Jacobian and Prym varieties, where the Gauss map can be described using canonical and Prym-canonical linear systems. This led to positive results for specific families, together with a study of how automorphisms, second fundamental forms, and curvature constrain the fibres of the tangent Gauss map.

A recent counterexample shows that the proposed symmetry explanation fails even for a simple ambient abelian variety. Its construction uses a non-hyperelliptic curve X of genus four with a theta characteristic L having two independent sections. A simple three-dimensional quotient A of its Jacobian is chosen so that the invariant one-forms pulled back from A form the space of products of sections of L. The induced tangent Gauss map factors as a degree-three map to the projective line followed by the quadratic Veronese embedding. Its image curve in A has no nontrivial preserving ambient automorphism with scalar differential. Thus the geometry of the relevant linear system can produce repeated tangent directions without the predicted symmetry.

The project will use this example to develop more precise questions about the possible degrees and factorizations of tangent Gauss maps. I will study how subspaces of canonical forms arising from abelian quotients interact with pencils on curves and with the rational Hodge structures defining those quotients. Initial attention will be given to low-genus curves and to families accessible through Prym constructions. The aim is to find sufficient conditions for generic injectivity and to identify mechanisms producing higher degree. The role of ramification and the variation of the second fundamental form along Gauss fibres will also be examined.

These questions have consequences for conormal monodromy. A factorization of the tangent Gauss map can impose a block structure on a general conormal fibre. I will determine this structure in explicit families and study how it affects the irreducible components of subvarieties formed by weighted sums of fibre points. Existing formulas for Chern-Mather classes in the full symmetric and full signed symmetric cases provide a basis for computing their dimensions and homology classes.

The broader comparison will include the Schoen-surface example, where two half-spin cycles have unequal homology classes despite arising from closely related monodromy orbits. Together, these investigations aim to clarify which geometric conclusions follow from monodromy and which require additional information about the embedding. The expected results are refined injectivity and monodromy criteria, explicit exceptional families, and applications to the geometry of convolution constructions on abelian varieties.